Spring 2026 Honours Project Report

Circuits of Salt: Commodity Networks and the Politics of Trade in the Western Himalayas

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1 Introduction

This honours project investigates the political economy of salt and multi-commodity trade networks in the Western Himalayas, particularly across Ladakh, Western Tibet, Changthang, Rupshu, and adjoining frontier regions. Building upon earlier archival and computational work, the Spring 2026 phase of the project focused on understanding how salt functioned not merely as a commodity, but as an infrastructure of governance shaping sovereignty, taxation, mobility, and frontier authority across Himalayan regions.

The project examines how precolonial systems of exchange operated through kinship obligations, reciprocity, pastoral mobility, and customary rights, and how these systems were progressively transformed under colonial rule into bureaucratically regulated, legible, and taxable economic structures. In this framework, salt routes are understood as the structural backbone of wider multi-commodity exchange systems involving pashm, wool, barley, grain, borax, tea, carpets, and textiles.

A major intellectual development during Spring 2026 was the integration of theoretical frameworks drawn from the works of Karl Polanyi, James C. Scott, E. P. Thompson, and Prasannan Parthasarathi. These frameworks enabled the project to situate Himalayan salt circulation within broader debates on embedded economies, moral economy, colonial market formation, labour autonomy, dispossession, and state legibility.

Simultaneously, the computational dimension of the project was significantly expanded. Earlier Named Entity Recognition (NER) and relation extraction pipelines were consolidated into a more systematic archival-to-network workflow capable of generating structured relational datasets from historical corpora. The project increasingly aims to combine computational text analysis with historical interpretation in order to reconstruct political and economic relationships embedded within Himalayan trade systems.

2 Research Objectives

The Spring 2026 phase of the project focused on consolidating the historical, theoretical, and computational foundations established during earlier stages of the honours research. The primary research objectives were centred around three interconnected themes.

The first objective was to engage extensively with the social science and historical literature surrounding salt, particularly within the context of the Western Himalayas. This involved studying anthropological, historical, and political-economic scholarship examining salt as a commodity, a medium of exchange, a subsistence resource, and a mechanism of governance. The project explored how salt shaped mobility, sovereignty, pastoral organisation, barter systems, and frontier relations across Himalayan regions.

The second objective was to investigate the broader history of commodities and exchange systems in the Western Himalayas. Rather than examining salt in isolation, the project aimed to situate it within wider multi-commodity networks involving pashm, wool, grain, tea, borax, textiles, carpets, and pastoral trade systems. Particular emphasis was placed on understanding embedded economies, reciprocity-based exchange, seasonal migration, customary rights, and the interaction between pastoral, agrarian, monastic, and frontier communities.

The third objective was to study the political economy of salt in Colonial North India, especially the transformation of Himalayan exchange systems under colonial rule. The project examined how colonial governance introduced monopolies, taxation regimes, permits, licensing systems, frontier surveillance, and territorial legibility into previously fluid and kinship-regulated exchange networks. Through the theoretical frameworks of Karl Polanyi, James C. Scott, and E. P. Thompson, the research analysed how embedded and subsistence-oriented economies were progressively reorganised into bureaucratically controlled market structures.

An additional objective during the Spring 2026 phase was to continue and refine the computational model work initiated during the summer period. This included improving the archival processing pipeline, expanding Named Entity Recognition workflows, refining co-occurrence analysis, and developing LLM-assisted relation extraction systems capable of generating structured relational CSV datasets and preliminary knowledge graphs from historical corpora.

3 Methodology

The project employs an interdisciplinary methodology combining archival research, political economy, computational text analysis, and network-oriented historical interpretation. The entire end-to-end processing architecture is managed automatically by a centralized master orchestrator (`pipeline.py`).

A large corpus of historical material was assembled from colonial gazetteers, treaties, ethnographies, administrative reports, settlement records, Foreign Department correspondence, and Himalayan trade studies. These materials were digitized and processed into machine-readable textual corpora using automated extraction tools built with Python and the PyMuPDF library, saving the output as raw text chunks in a shared `corpus/` directory.

Named Entity Recognition was conducted using the TextRazor API, which simultaneously leverages entity linking to identify distinct historical entities and maps them to Freebase topic hierarchies. The system processes the text in localized windows, matching actors and commodities against broader categories like `/food/food`, `/location/region`, and `/government/governmental_jurisdiction`. This structural alignment generates a detailed foundational dataset stored in `ner_results.txt`.

Topic aggregation and thematic filtering were then performed through regular-expression based parsing to build an alphabetized dictionary of unique classifications (`unique_topics.txt`). Rather than relying purely on automated sorting, the project uses a intentional **Human-in-the-Loop** checkpoint. The researcher reviews this list to strip away noise and select historically meaningful conceptual filters saved into `Selected_Topics.txt`, which explicitly shapes all downstream passes.

Co-occurrence analysis was implemented using a sliding ±50-word contextual window around selected anchor entities such as salt, pashm, taxation, monasteries, and frontier settlements. This allowed the project to identify recurring associations between commodities, trade routes, institutions, and political structures within historical texts. The frequency counts and localized textual evidence snippets are written to `entity_cooccurrences.txt`.

The relation extraction pipeline was expanded significantly during Spring 2026. To bypass computational bottlenecks, the co-occurrence file filters and pre-selects candidate entity pairs. Historical passages containing these pairs are then processed using a locally deployed Llama 3 model running through Ollama to maintain strict data privacy. The model operates under a highly constrained, domain-specific taxonomy of allowed directional relationships:

- **Economic Interaction:** *trades_with, exchanges_for, extracts_from, transports_via*
- **Governance & Friction:** *taxes, regulates, governs, controls, disputes, licenses,*

monopolizes

- **Structural Networks:** *supplies, depends_on, connects_to, migrates_through, administers, negotiates_with*

Identical triples are aggregated and evaluated by frequency of occurrence, generating a robust tabular graph layout in `weighted_knowledge_graph.csv`.

To ensure network hygiene as new texts are ingested, an entity type correction subroutine cross-references newly found nodes with previous records, maintaining consistent categories across the global index. Ambiguous node types are flagged for human validation via `entity_type_review.csv`, preserving custom overrides mapped in `config.py`, before outputting a harmonized network blueprint (`cleaned_entities.json` and `cleaned_network.graphml`).

Finally, the network is rendered into a dynamic, browser-based visual layout via `network_visualization` The interface represents entities as specialized nodes color-coded by category (e.g., Commodities, Locations, Groups, People), scales node dimensions based on network centrality, and sets edge thicknesses according to relationship frequency weights to enable interactive macro- and micro-level exploratory analysis.

4 Progress and Work Completed

Substantial progress was achieved during the Spring 2026 phase across theoretical, computational, archival, and research communication dimensions.

A major portion of the semester was devoted to strengthening the theoretical framework of the project. Extensive reading and note compilation were carried out on the works of Polanyi, Scott, Thompson, and Parthasarathi. These readings helped situate Himalayan salt trade within broader discussions concerning embedded economies, colonial capitalism, labour autonomy, dispossession, and political reorganisation. The project increasingly developed toward a comparative understanding of colonial market formation across both Himalayan trade systems and South Indian textile economies.

Detailed analytical notes and bibliographic summaries were prepared for CTR-II, focusing on themes such as reciprocity, redistribution, substantivism, state simplification, moral economy, customary rights, labour discipline, and informal insurance systems. These notes later informed the writing of the CTR report and multiple presentation materials.

The computational infrastructure of the project was also expanded considerably, transitioning the theoretical workflow into an integrated software pipeline. The specific outputs generated by the automated data extraction and refinement pipeline over the course of the semester are structured as follows:

Co-occurrence analysis and relation extraction produced preliminary evidence supporting

Table 1: Pipeline Architecture Data Deliverables

Output File	**Research Function & Description**
`ner_results.txt`	Comprehensive named entities and Freebase topics with confidence metri
`entity_cooccurrences.txt`	Proximity metrics, window frequencies, and localized text fragments
`weighted_knowledge_graph.csv`	Extracted Subject–Relation–Object triples with factual justification
`cleaned_entities.json`	Master index of validated, deduplicated entity types across texts
`cleaned_network.graphml`	Standard structural graph schema compiled for formal network metrics
`network_visualization.html`	Interactive browser interface for multi-layered network exploration
`entity_type_review.csv`	Targeted workspace for historical validation of ambiguous nodes

the hypothesis that salt functioned as the central infrastructural node connecting multiple commodities and regional trade systems. Computational outputs indicated strong relational clustering between salt extraction sites, pastoral communities, administrative institutions, and taxation systems.

The Rupshu and Tso Kar basin continued to serve as the central case study for the project. Archival and ethnographic analysis documented seasonal extraction systems, chu lag labour organisation, kotwal supervision, territorial disputes, and the integration of salt with pastoral and barter economies. These findings reinforced the argument that control over salt routes constituted a form of political and territorial authority.

Several research dissemination activities were completed during the semester. These included the preparation of posters and presentations for the R&D Showcase, the drafting of a CTR paper titled *Salt, Subsistence, and Sovereignty: The Political Economy of the Himalayan Salt Trade through Scott, Polanyi, and Thompson*, and the preparation of one-page summaries and conceptual abstracts. The project also produced presentation materials connecting Coromandel textile labour systems with Himalayan salt circulation through shared themes of embedded economies and colonial disembedding.

5 Future Work

Future work will focus on expanding both the analytical depth and computational sophistication of the project.

The first priority will be the continued refinement of the weighted relational graph and knowledge network generated through LLM-based extraction. Additional validation and manual correction of Subject–Relation–Object triples will be necessary to improve historical accuracy, especially for region-specific terminology, OCR inconsistencies, and ambiguous colonial spellings.

The second priority will be the integration of temporal and spatial analysis into the network framework. This includes investigating how trade routes, taxation structures, and commodity relationships changed across different colonial and postcolonial periods.

Spatial visualisation of routes such as the Rupshu–Tso Kar–Leh axis and Shipki corridor may be incorporated into later stages of the project.

Third, the project will continue to develop comparative frameworks connecting Himalayan salt circulation with broader histories of colonial capitalism, labour reorganisation, and market formation. The parallels drawn between Coromandel textile systems and Himalayan exchange networks will be expanded further.

Fourth, additional archival material, particularly Foreign Department records, salt administration reports, frontier surveys, and settlement documentation, will be incorporated into the corpus. These materials will strengthen the sections dealing with colonial governance, territorial legibility, licensing systems, and frontier surveillance.

Finally, future writing will increasingly focus on synthesising computational outputs with historical interpretation. Rather than treating NLP analysis as a separate technical layer, the project aims to use network modelling and relation extraction as evidence for larger arguments regarding embedded economies, political authority, and colonial transformation.

6 Conclusion

The Spring 2026 phase of the honours project significantly advanced both the theoretical and computational dimensions of the study. The project has increasingly moved beyond a narrowly defined commodity history toward a broader investigation of political economy, frontier governance, embedded exchange systems, and colonial market formation in the Western Himalayas.

The research demonstrates that salt routes were not merely channels of commodity transport, but infrastructures through which sovereignty, mobility, taxation, and territorial control were negotiated and exercised. Precolonial systems organised through reciprocity, kinship, redistribution, and customary authority were progressively transformed into legible and bureaucratically regulated economic systems under colonial rule.

By integrating archival analysis with computational text processing, co-occurrence analysis, and relation extraction, the project has developed a framework capable of reconstructing historical trade networks from large textual corpora. The work completed during Spring 2026 establishes a strong foundation for future research involving network modelling, spatial analysis, and comparative studies of colonial political economy across South Asia.